Nguyễn Đại Đồng

Ho Chi Minh City, Viet Nam   dong.nguyendai29k22@hcmut.edu.vn   0899 478 570
dong-nguyen-hcmut   Noname-29-lnin

Education

VNU - Ho Chi Minh City University of Technology *Oct 2022 – Present*
Bachelor of Science in Electronics Engineering

- **GPA:** 3.0/4.0 (This link)
- **Project 1:** Design and Implementation Viterbi Decoder Algorithm on FPGA.

Research Interest

Computer Architecture, RISC-V Processors, Hardware Accelerato, Digital IC Design.

Technical Skill

- Programming Language & HDL:
  + Proficient: C, C++, SystemVerilog, Verilog.
  + Scripting & Documentation: LaTeX, Bash.
  + Tools: Vim, GIT, Makefile.
- Development Environments:
  + Operating Systems: Linux (Ubuntu, CentOS), Windows.
  + FPGA Tools: Intel Quartus, QuestaSim, ModelSim, Verilator, Virtuoso, Xcelium.
- Hardware Skills:
  + Digital Design: RTL development, Verification.
  + PCB Design: KiCad, Alitium (Basic).
  + Lab Equipment: Oscilloscope, Logic Analyzers.

Projects

Design and Implementation of a Power-Efficient Viterbi Encoding and Decoding Architecture on FPGA: From Theory to Practical Application *Noname-29-lnin/Viterbi_decoder*

- This project involves the design and implementation of a **Viterbi decoder** for convolutional codes with parameters $(3, 1, 2)$ (constraint length $K = 3$, code rate $1/2$) on an FPGA. The Viterbi algorithm is optimized for error correction in noisy communication channels, leveraging trellis-based path metrics and survivor path selection. The decoder is structured using key functional blocks, including the *Branch Metric Unit (BMU)*, *Add-Compare-Select Unit (ACSU)*, *Path Metric Unit (PMU)*, and *Survivor Path Memory Unit (SPMU)*.
- To test the decoder, additional modules such as a *Parallel-In-Serial-Out (PISO)* register, *Serial-In-Parallel-Out (SIPO)* register, and *LCD IP core* (for hardware-based LCD control) are integrated.

These components enable practical verification of the Viterbi decoder's functionality. The final implementation is deployed on a *DE2 development kit*, where the design is synthesized and programmed onto the FPGA for real-world validation.

Design a combination circuit: A 1-bit Full Adder

- Use Virtuoso to layout a 1-bit Full Adder circuit using different approaches, such as the basic logic circuit of the FA: $S = A \oplus B \oplus C_{in}$, $C_{out} = (A \oplus B)C_{in} + AB$, as well as the 28T Full Adder structure and the 24Y Full Adder using Mirro adder.

Design of a Single Cycle RISC-V Processor *Noname-29-lnin/Single-Cycle-RISC-V*

- Asingle-cycle implementation of the RISC-V RV32I instruction set architecture (ISA), designed from scratch in SystemVerilog. This project demonstrates the core principles of CPU microarchitecture, including instruction fetch, decode, execution, memory access, and writeback stages—all completed within a single clock cycle.

Design Adder 32-bit using Verilog *Noname-29-lnin/Adder_32bit*

- Acomprehensive implementation and comparison of four classic 32-bit adder architectures in Verilog, analyzing the trade-offs between speed, area, and power consumption. Implementation Architecture:
  - Carry Ripple Adder (CRA)
  - Carry Lookahead Adder (CLA)
  - Carry Select Adder (CSA)
  - Carry Skip Adder (CSKA)

Design of a DC/AC Voltage measurement and waveform display device using STM32F1xx *Noname-29-lnin/OSC_project*

- An embedded measurement system using STM32F1xx microcontroller for accurate voltage acquisition and real-time waveform display on PC, implementing core functionalities of a basic oscilloscope.
- Tools & Lauguage:
  - Tool used: KiCad, Keil-C, STM32CubeMX, Proteus 8, ST-Link programmer.
  - Language: C, Python.